SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2024-2025

Program	:	BTI/B.Tech/MBA Tech	Year	:	IV /II
Stream	:	Data Science	Semester	:	VIII/IV
Subject	:	Stochastic Process & Applications (STPA)	Time	:	3 hrs (10:00AM to 1:00PM)
Date	:	28 April 2025	No. of Pages	:	3
Marks	:	100			

Final Examination 2024-25/ Re-Examination 2022-23/2023-24

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Solve Total FIVE questions.
2) Question number 01 is compulsory.
3) All questions carry equal marks.
4) **Answer to each new question to be started on a fresh page.**
5) **Figures in brackets on the right-hand side indicate full marks.**
6) **Assume Suitable data if necessary.**
7) **Scientific Calculator & Statistical Table is allowed to use.**

Q1.	**Answer in Detail**	[20]
a.	Define first and second-order statistics of a random process.	[5]
b.	The lifetime of certain brand of electric bulb may be considered a random variable with mean 1200 and SD 250 hours. Using CLT to find the probability that the average lifetime of 60 bulbs exceeds 1250 hours.	[5]
c.	Consider a Markov Chain with states $\{0, 1\}$ and the transition probability matrix is $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Is the state 0 is periodic, if yes, find the period?	[5]
d.	If the autocorrelation of a random process is given by $R(\tau) = \begin{cases} \lvert\tau\rvert & \lvert\tau\rvert \le 1 \\ 0 & \lvert\tau\rvert > 1 \end{cases}$ then find the spectral density function $S_{xx}(\omega)$.	[5]
Q2.	**Answer in Detail**	[20]
a.	Find the mean, mean square and variance of a random process whose autocorrelation function is given by $R_{XX}(\tau) = 18 + \dfrac{2}{(6+\tau^2)}$	[6]
b.	Consider the outcomes of the tossing of fair coins are described by the following function of time $X(t) = \begin{cases} \sin \pi t, & \text{If head appears.} \\ 2t, & \text{If tail appears.} \end{cases}$ Where $t > 0$. Describe the random process.	[6]
c.	If X denotes the outcome when a fair die is tossed. Then obtain the MGF $M_X(t)$ and henceforth figure out the corresponding mean and variance of the distribution.	[8]
Q3.	**Answer in Detail**	[20]
a.	Suppose that the average number of telephone calls arriving at a police station is 30 calls per hour. What is the probability that (i) No call will arrive in a 3-minute period. (ii) More than 4 calls will arrive in a 5-minute period?	[6]
b.	Two random processes are defined by $X(t) = \cos(\omega t + \Theta)$ and $Y(t) = \sin(\omega t + \Theta)$, where ω is constant and $\Theta \sim U[-\pi, \pi]$. Find the cross correlation $R_{XY}(\tau)$.	[6]
c.	Show that the random process given by $X(t) = X_1(t)\cos t + X_2(t)\sin t$, where X_1, X_2 are independent discrete random variables assuming values -1 and 2 with probabilities $\frac{2}{3}$ and $\frac{1}{3}$ is wide sense stationary but not strict sense stationary.	[8]

Q4. Answer in Detail [20]

a. If a random process is given by $X(t) = A\cos(2\pi t + Y)$, Where A is a constant and Y is discrete random variable with $P(Y = 0) = P\left(Y = \frac{\pi}{2}\right) = \frac{1}{2}$, then find the mean and autocorrelation function. [6]

b. The transition probability matrix P of a Markov Chain $\{X_n\}$ where $n \in \mathbb{N}$, with the three states, numbered 1, 2, and 3 given by $P = \begin{bmatrix} 0.4 & 0.6 & 0.3 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.7 & 0.2 \end{bmatrix}$, with the initial distribution is $P^{(0)} = \begin{bmatrix} 0.6 & 0.3 & 0.1 \end{bmatrix}$

Then find the probabilities of the following states:

(a) If today is Wednesday and the weather is in state 2 what is the probability that the weather is in state 3 on Thursday and state 1 on Friday?

(b) What is the probability that the weather on Friday is in state 1, given that it is in state 2 on Wednesday? [6]

c. A continuous function of a random variable X has the p.d.f $f(x) = kx^2e^{-x}$ where $x \geq 0$. Then find

(i) k,

(ii) Mean

(iii) Variance. [illegible]

Q5. Answer in Detail [20]

a. If the power spectral density $S_{xx}(\omega)$ of a random process is given by

$$S_{xx}(\omega) = \frac{10\omega^2 + 35}{\omega^4 + 13\omega^2 + 36}$$

then find the autocorrelation function $R_{xx}(\tau)$ and hence obtain the average power of the process. [6]

b. If $\{X(t)\}$ is a Gaussian process with $\mu(t) = 10$ and the cross correlation function is $C(t_1, t_2) = 4e^{-0.2|t_1 - t_2|}$ Then find $P(X(8) \leq 3)$ and $P(X(8) - X(3)) \leq 3$. [6]

c. Suppose in a gambling game a gambler wins and losses rupees 1 with probability 0.45 and 0.55 respectively at any stage. He starts the game with 1 rupee and quits it when either he has earned rupees 4 or he is bankrupt. Define the process as a Markov chain and write the transition matrix. Also obtain the transition probability of sequences $\{12343210\}$ and $\{234345\}$. [8]

Q6. Answer in Detail [20]

a. Consider the DRV X is associated with the following PMF

X	-1	0	1
$P_X(x)$	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{8}$

Then evaluate $P\{|X - \mu| \geq 2\sigma\}$ and compare the result with the upper bound of Chebyshev's inequality. [6]

b. The autocorrelation function of a stationary random process is given by $R_X(\tau) = Ae^{-\alpha|\tau|}$. Find the second moment of the random variable $Y = X(6) - X(3)$. [6]

c. Prove that the random process $X(t) = A\cos(\omega t + \Theta)$ is mean ergodic and correlation-ergodic, where A and ω are constants and $\Theta \sim U[0, 2\pi]$. [8]

Q7. Answer in Detail [20]

a. A random variable X has the following probability distribution [6]

X	0	1	2	3	4	5	6	7
$P_X(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k$	$7k^2 + k$

Then find

(i) k

(ii) $P(0 < X < 5)$

(iii) The CDF of X.

b. A sine wave random process is given by $X(t) = A\sin\omega_0 t$ where $A \sim U[-1, 1]$ and ω_0 is a constant. Check whether $\{X(t)\}$ is WSS. [6]

c. Find the spectral density function $S_{xx}(\omega)$ where $R(\tau) = e^{-a|\tau|}\cos b\tau$. [8]

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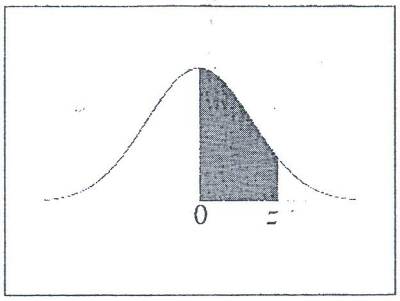
Standard Normal Distribution Table



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.1	.4990	.4991	.4991	.4991	.4992	.4992	.4992	.4992	.4993	.4993
3.2	.4993	.4993	.4994	.4994	.4994	.4994	.4994	.4995	.4995	.4995
3.3	.4995	.4995	.4995	.4996	.4996	.4996	.4996	.4996	.4996	.4997
3.4	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4998
3.5	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998